

Dhiraj Kumar Sah

+977-9815310840 | eyemdheeraj436@gmail.com | dhirajsah99.com.np | Dhiraj Sah | DhirajSah736

Itahari-17, Koshi - 56705, Nepal

SUMMARY

Computer Science graduate and aspiring AI governance researcher with a strong foundation in web technologies, AI applications, and ethical technology development. Experienced in community outreach and education, with a deep commitment to advancing responsible, culturally informed, and globally relevant AI systems. Dedicated to bridging technical innovation with sociocultural and policy considerations for real-world impact.

EDUCATION

- Chandigarh University** 2020 - 2024
Bachelor of Engineering - Computer Science and Engineering
Punjab, India
◦ GPA: 8.16/10
- Delhi Public School-BPKIHS (CBSE)** 2017-2019
Intermediate
Dharan, Nepal
◦ Grade: 68.16%
- Bal Bikas English Boarding School (NEB)** 2004-2017
Matriculation
Itahari, Nepal
◦ GPA: 3.15/4

PROJECTS

- MyFitCoach: AI-powered Fitness App [WEB APPLICATION]** May 2025 - Jun 2025
Tools: [React, Vite, Supabase, Stripe, Tavus AI, Tailwind CSS] [🔗] [🌐]
◦ Integrated AI-driven coaching with privacy-first design principles and equitable access considerations, aligning with responsible AI deployment practices.
◦ Developed Supabase Edge Functions to handle Stripe webhooks for subscription management, improving payment reliability by 40%.
◦ Designed premium access logic with free trials and promotional codes, increasing user retention and monetization efficiency.
◦ Created a modern, responsive UI with Tailwind and animations, boosting mobile usability by 50%.
- Sorting Visualizer: [WEB APPLICATION]** May 2024 - Aug 2024
Tools: [React, Vite, Tailwind CSS, Framer Motion] [🔗] [🌐]
◦ Developed a tool visualizing sorting algorithms (Bubble, Selection, Insertion, Merge, Quick, Heap), enhancing user understanding by 200%.
◦ Focused on improving computational literacy by making algorithmic processes intuitive for learners globally.
◦ Designed an interactive UI, increasing user engagement by 40%.
◦ Optimized DOM to handle 1,000 data points, boosting visualization speed by 25%.
- WeatherPulse-Weather forecasting App: [WEB APPLICATION]** Dec 2023 - Apr 2024
Tools: [React, vite, Typescript, WeatherAPI] [🔗] [🌐]
◦ Designed for accessibility and inclusivity across diverse user groups and geographies.
◦ Engineered a client-side application for delivering real-time meteorological data via dynamic city-based queries, leveraging RESTful APIs from WeatherAPI.com.
◦ Implemented granular data rendering for temperature, humidity, wind speed, and atmospheric conditions with semantic UI components and iconography.
◦ Designed and deployed a responsive architecture using React hooks and Tailwind CSS utility classes, ensuring consistent UX across varied viewport sizes.

RESEARCH PAPERS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [S.1] Dhiraj Kumar Sah (2025). **FedFit: A Federated Multi-Modal Deep Learning Framework for Privacy-Preserving Intelligent Fitness Coaching**. Manuscript submitted for publication at *ASIANCON 2025*.
(Addresses privacy-preserving AI frameworks relevant to global governance and ethical deployment.)
- [C.1] Dhiraj Kumar Sah (2024). **Designing a Responsive and Accessible Blood Bank Management Web Application Using MERN Stack**. Accepted for oral/poster presentation at *ICRCET-2024: International Conference on Recent Challenges in Engineering and Technology*, Bangalore, India, April 23–24, 2024. [Letter of Acceptance]

SKILLS

- **Programming & Frameworks:** C++, Python, JavaScript (React, Node.js, Express.js), HTML5, CSS3
- **AI & Data Systems:** Machine Learning, Federated Learning, Privacy-Preserving AI, Data Engineering (Spark), MongoDB, MySQL, NoSQL
- **Tools & Platforms:** Supabase, Vite, Tailwind CSS, Git, GitHub, AWS, Stripe API
- **Governance & Policy Literacy:** Responsible AI Deployment, Ethical Technology Design, Global Digital Inclusion Awareness
- **Professional Skills:** Cross-cultural Communication, Leadership, Community Outreach, Team Collaboration

VOLUNTEER EXPERIENCE

- **Volunteer Tech Tutor** Aug 2024 - Oct 2024
Bal Bikas English Boarding School 
 - Advocated for equitable technology education in underserved communities, aligning with global digital inclusion efforts.
 - Developed and delivered interactive, hands-on technology lessons tailored to secondary students, fostering critical digital literacy skills.
 - Facilitated an activity-driven learning environment that significantly increased student engagement and comprehension of complex technical concepts.
 - Cultivated strong communication and instructional skills by adapting content to diverse learning styles and encouraging student curiosity and confidence.

CERTIFICATIONS

- **Machine Learning for Computer Vision** Aug 2023
MathWorks | Coursera
Credential ID: RNTVHAJD6CQC
- **Data Engineering and Machine Learning using Spark** May 2023
IBM | Coursera
Credential ID: TKX9Y78JR8QJ
- **Software Testing** Oct 2022
NPTEL
Credential ID: NPTEL22CS61S6467039810052910
- **Java Programming: Solving Problems with Software** May 2022
Duke University | Coursera
Credential ID: BRVED96VH6BJ
- **Programming Foundations with JavaScript, HTML and CSS** May 2022
Duke University | Coursera
Credential ID: WMVPYFUKCZKG
- **Programming for Everybody (Getting Started with Python)** May 2022
University of Michigan | Coursera
Credential ID: HT5B2E9VNLFS

HACKATHONS AND COMPETITIONS

- **World’s Largest Hackathon – Bolt.new AI** May 2025 – July 2025
Bolt.new AI
 - Designed and deployed *MyFitCoach*, an AI-powered fitness coaching platform integrating Tavus AI and privacy-first user interactions.
 - Focused on equitable AI access and ethical deployment of generative video coaching, aligning with global AI governance principles.
 - Developed premium access workflows balancing user experience with data privacy considerations.
- **Google Maps Platform Awards** June 2025 – Present
Google Developers
 - Developing a project leveraging Google Maps APIs for sustainable urban mobility and equitable resource distribution.
 - Emphasizing AI-enabled geospatial insights for governance and ethical deployment in smart cities and disaster management.